

**A REPORT
ON**

**A TWO-STAGE SUMMARIZATION PIPELINE FOR
NEWS ARTICLES:**

**EXTRACTION-GUIDED ABSTRACTIVE
GENERATION**

BY

Puthineedi Venkata Sai Charan

2024AA05606

AT

Opentext

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

August 2026

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M.Tech. Artificial Intelligence and Machine Learning

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AT

Opentext

Under the Supervision of

Veeraswamy Ponnuru, Lead Quality Assurance Engineer

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

August 2026

Declaration

I, Puthineedi Venkata Sai Charan (ID: 2024AA05606), declare that the dissertation titled “A Two-Stage Summarisation Pipeline for News Articles: Extraction-Guided Abstractive Generation with Evidence Retrieval, NLI-based Factual Verification, and a Difficulty-Aware Safety Switch”, submitted to BITS Pilani, Work Integrated Learning Programmes Division, in partial fulfilment of the requirements for the degree of Master of Technology in Artificial Intelligence and Machine Learning, is a record of original work carried out by me under the supervision of Veeraswamy Ponnuru, Lead Quality Assurance Engineer, Opentext.

I further declare that this work, in part or in full, has not been submitted for the award of any other degree or diploma of any university or institution, and that all sources of information used have been duly acknowledged.

Place: Hyderabad

Date: 01 August 2026

P.V.Sai Charan
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I acknowledge the open-source community behind Hugging Face Transformers, the maintainers of the CNN/DailyMail benchmark, and the authors of BART, PEGASUS, DeBERTa and Sentence-Transformers, whose freely available models made this research possible within the compute budget of a part-time programme. I also acknowledge Google LLC for the Colab compute resources used to run the fine-tuning and evaluation experiments.

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Abstract

Abstractive summarisation with pre-trained transformers produces fluent output but is prone to hallucination, and single-stage pipelines offer no mechanism to detect or suppress unsupported claims. This dissertation implements and evaluates a four-stage pipeline — hybrid BM25 and sentence-embedding evidence retrieval, Best-of-N abstractive generation with a fine-tuned BART model, DeBERTa-based NLI factual verification, and a preference reranker with a novel Difficulty-Aware Safety Switch — evaluated on CNN/DailyMail. At a 50-sample ablation scale, hybrid retrieval and TextRank achieve statistically comparable ROUGE, both clearly ahead of BM25-only; increasing the candidate count in Best-of-N generation improves ROUGE with diminishing and non-monotonic returns beyond N=3–5; and the NLI verifier improves ROUGE-1/2/L and BERTScore over an unverified variant while adding acceptable latency. The full pipeline reaches an NLI factual-consistency score above 99% on a 300-article evaluation with zero triggered extractive fallbacks.

*P.V.Sai Charan**Veeraswamy Ponnuru*

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Chapter 1 Introduction

1.1 Background and Motivation

The volume of news content published online each day has grown well past what any individual reader, editor, or downstream system can review in full. Automatic summarisation is one of the more direct responses to this problem: it produces a shorter representation of an article that preserves the information a reader actually needs, cutting the time spent reading without discarding the substance of the source.

Two broad families of technique have shaped this area. Extractive methods select and concatenate sentences taken directly from the source document; because every output token already appeared in the input, they are structurally incapable of inventing facts, but they tend to read as a disjointed sequence of lifted sentences rather than a coherently written summary. Abstractive methods, built on pre-trained sequence-to-sequence transformers such as BART (Lewis et al., 2020) and PEGASUS (Zhang et al., 2020), generate new text and can compress, paraphrase, and fuse information across sentences — but nothing in their training objective prevents them from generating a plausible-sounding claim that the source article never made.

CNN/DailyMail (Hermann et al., 2015; Nallapati et al., 2016) has been the standard English news-summarisation benchmark for close to a decade, and most published results on it are reported using ROUGE (Lin, 2004), an n-gram overlap metric against human-written reference summaries. ROUGE is a reasonable proxy for content selection and fluency, but it has no way of penalising a summary that reads well and overlaps with the reference yet states a fact the source article does not support. This gap between what ROUGE measures and what a reader actually needs — a summary they can trust — is the starting point for this dissertation.

This work is motivated by a narrower observation: no single openly available system combines hybrid evidence retrieval, multi-candidate abstractive generation, sentence-level NLI-based verification, and an adaptive safety fallback within one CNN/DailyMail pipeline. The system described in this dissertation is built to close that specific gap, and the ablation studies in Chapter 6 are designed to test, rather than assume, that each added component earns its place.

1.2 Research Problem

Single-stage abstractive pipelines for news summarisation share three limitations that motivate the design of this project:

- Truncation bias — encoder token limits (typically 512–1024 tokens) force truncation of longer articles, so content appearing later in the source is simply unavailable to the generator.
- Hallucination risk — a transformer decoder can produce a fluent, well-formed sentence that is not supported by the source, with no built-in mechanism to flag or suppress it.
- Fixed-threshold brittleness — where a factual-confidence fallback exists at all, it typically applies one threshold to every article regardless of how difficult that particular article is to summarise, which either triggers unnecessary fallbacks on easy articles or misses genuine hallucinations on hard ones.

Addressing all three within the compute budget realistically available to a part-time, work-integrated postgraduate programme is the central problem this dissertation takes on.

1.3 Objectives

- Design and implement a four-stage pipeline: evidence retrieval, Best-of-N abstractive generation, NLI-based factual verification, and preference reranking with extractive fallback.
- Fine-tune BART-large on CNN/DailyMail within Colab-tier GPU constraints and compare it against zero-shot and pre-trained baselines.
- Design and evaluate a Difficulty-Aware Safety Switch that adapts the factual-confidence fallback threshold to per-article difficulty rather than applying one fixed value.
- Conduct systematic ablation studies isolating the contribution of retrieval strategy, candidate count, and the verifier stage, at a sample size sufficient to distinguish a real effect from ablation noise.
- Assess multilingual feasibility by extending the pipeline to Hindi summarisation via mBART and XL-Sum.

1.4 Research Questions

RQ	Question
RQ1	Does hybrid BM25 + sentence-embedding retrieval produce better evidence context for BART than lexical or semantic retrieval alone, measured by downstream ROUGE?
RQ2	How does increasing the candidate count N in Best-of-N generation affect ROUGE and inference latency?
RQ3	Does the NLI verifier improve factual consistency and/or ROUGE over unverified generation, at a sample size large enough to be more than directional?
RQ4	Does the Difficulty-Aware Safety Switch adapt the fallback threshold sensibly to article difficulty without an excessive fallback rate?
RQ5	Is the pipeline architecture generalisable to a low-resource multilingual setting?

1.5 Scope and Constraints

- Primary dataset: CNN/DailyMail 3.0.0.

- Generator model: facebook/bart-large-cnn, fine-tuned for two additional epochs on 20,000 training samples — a deliberate constraint given Colab free/pro-tier GPU time, not a claim of exhaustive training.
- Ablation sample size: 50 articles per condition (revised upward from an initial $n=15$ development pass — see Chapter 6 for why the larger sample changed several conclusions); primary system comparison uses 300 articles.
- A human-evaluation protocol is designed and documented in the evaluation harness; full-scale annotation is deferred as future work due to annotator-availability constraints.
- Multilingual study: a Hindi XL-Sum feasibility check only, not a claim of production-grade multilingual coverage.

1.6 Contributions

- A modular, open-source four-stage summarisation pipeline (retrieval → Best-of-N generation → NLI verification → preference reranking) evaluated systematically on CNN/DailyMail.
- A Difficulty-Aware Safety Switch — an inference-time mechanism that replaces a single fixed factual-fallback threshold with one conditioned on article-level difficulty signals (length, entity density, retrieval uncertainty).
- Ablation evidence at a sample size ($n=50$) sufficient to catch cases where a smaller pilot's conclusion did not hold up — Chapter 6 documents two such reversals honestly rather than silently keeping the earlier claim.
- A Hindi feasibility extension demonstrating that the architecture is not English-specific.
- A reproducible codebase: training notebooks, an evaluation harness, a Streamlit demo, and a single configuration module.

1.7 Report Organisation

Chapter 2 reviews the literature underlying each pipeline stage. Chapter 3 states the problem and objectives formally. Chapter 4 describes the system architecture. Chapter 5 covers the implementation. Chapter 6 presents the experimental evaluation, including the revised $n=50$ ablation results. Chapter 7 discusses the findings and limitations. Chapter 8 concludes the dissertation.

Chapter 2 Literature Review

2.1 Extractive Summarisation

TextRank (Mihalcea & Tarau, 2004) represents each sentence as a node in a graph, with edges weighted by lexical similarity, and ranks sentences by eigenvector centrality. It requires no training data and remains a competitive baseline on news text because journalistic writing tends to front-load the most important information.

Later neural extractive methods reframed sentence selection as a scoring or classification problem. SummaRuNNer (Nallapati, Zhai, & Zhou, 2017) scores each sentence in context with a recurrent encoder, jointly modelling salience, novelty, and position. BertSum (Liu & Lapata, 2019) replaces the recurrent encoder with a pre-trained transformer and improves extractive ROUGE on CNN/DailyMail while, like all extractive methods, guaranteeing that every output token appears verbatim in the source.

In this dissertation, TextRank serves two roles: as a standalone extractive baseline, and as one of the four arms in the retrieval-strategy ablation (Chapter 6), letting the hybrid retriever be compared directly against a well-established, training-free method rather than only against its own components in isolation.

2.2 Abstractive Summarisation

Early sequence-to-sequence models with attention (Rush, Chopra, & Weston, 2015; Hermann et al., 2015) showed that an encoder-decoder architecture could produce fluent news summaries, but suffered from repetition and from omitting or inventing facts. The pointer-generator network (See, Liu, & Manning, 2017) partially addressed this by letting the decoder copy words directly from the source via a pointer mechanism, combined with a coverage loss to reduce repeated phrases.

Modern systems largely supersede copy mechanisms with large pre-trained transformers fine-tuned end to end. On CNN/DailyMail, a single-stage abstractive model encodes a (possibly truncated) article and decodes a summary in one pass — simple and fast, but with two structural weaknesses that motivate this dissertation's design: the encoder's token budget forces truncation of longer articles, and there is no explicit step that checks whether a generated claim is actually supported by the source.

2.3 Pre-trained Transformer Models

Three pre-trained sequence-to-sequence models are directly relevant to this work. BART (Lewis et al., 2020) is trained as a denoising autoencoder — span masking and sentence permutation corrupt the

input, and the model learns to reconstruct the original text. Fine-tuned on CNN/DailyMail, BART-large-cnn is reported to reach roughly ROUGE-2 21 on the public test set with a 1024-token encoder limit; in this project it is used in three roles — a zero-shot baseline, a pre-trained (not further fine-tuned) baseline, and the primary generator after fine-tuning on this project's own training split.

PEGASUS (Zhang et al., 2020) is pre-trained with a gap-sentence objective that specifically resembles the summarisation task: salient sentences are removed and the model learns to regenerate them from the remaining context. Because google/pegasus-cnn_dailymail was already trained by its authors on this exact benchmark, it is evaluated here zero-shot rather than fine-tuned again — re-training on data it has already seen would not add signal and would make the baseline comparison less fair, not more.

mBART (Liu et al., 2020) extends the BART pre-training objective to 25 languages via multilingual denoising. It is not CNN/DailyMail-specific, but it provides the model used for this dissertation's Hindi XL-Sum feasibility study, sharing the same underlying architecture as the English pipeline.

Table 2.1 — Comparison of Pre-trained Transformer Models

Model	Pretraining Objective	Max Tokens	CNN/DM ROUGE-2	Role
BART-large / BART-large-cnn	Denoising (span corruption + permutation)	1024	~21 (fine-tuned)	Primary generator
PEGASUS-cnn_dailymail	Gap-sentence generation	512	~21 (zero-shot)	Comparison baseline
mBART-large-cc25	Multilingual denoising	1024	Lower (cross-lingual)	Multilingual study

2.4 Factuality Evaluation

ROUGE (Lin, 2004) measures n-gram overlap with a reference summary and correlates reasonably with fluency and content selection, but cannot detect hallucination — a summary can score well on ROUGE while asserting something the source article never said. BERTScore (Zhang et al., 2020) improves on this by comparing contextual embeddings rather than raw n-grams, correlating better with human judgment, but it remains reference-based and does not check grounding in the source directly.

Reference-free factuality metrics increasingly rely on Natural Language Inference. SummaC (Laban et al., 2022) frames inconsistency detection as entailment between summary sentences and source chunks. The Factual Consistency Score (FCS) used in this dissertation follows the same idea at the sentence level, using a DeBERTa-v3 cross-encoder to produce an entailment probability for each summary sentence against the retrieved evidence pool.

Stress-testing work on reference-free factuality metrics (Mujahid, Wright, & Augenstein, 2026) found that metric reliability degrades as document length grows, but that NLI-based metrics remain among the more dependable automatic indicators for short-document settings such as CNN/DailyMail. This finding is the direct justification for using a sentence-level NLI verifier in Stage 3 of the proposed pipeline rather than a longer-document metric that the same study found less reliable at this scale.

2.5 Retrieval-Augmented Generation

Retrieval-Augmented Generation (Lewis et al., 2020) conditions a generator on retrieved evidence instead of the full document, improving grounding on knowledge-intensive tasks. For single-document summarisation, where there is no external query, self-retrieval — scoring each sentence of the article against the article's own content — is the standard adaptation.

Lexical retrievers such as BM25 (Robertson & Zaragoza, 2009) capture keyword overlap efficiently; dense retrievers using sentence embeddings capture semantic similarity that survives vocabulary mismatch. Combining both signals is a common way to make retrieval robust to cases where either signal alone is weak. The pipeline in this dissertation applies exactly this hybrid scoring to select the sentences that form both the generator's input and the evidence pool used later for NLI verification.

2.6 Research Gap and Contribution

Individually, prior work addresses extractive faithfulness, abstractive fluency, or post-hoc factuality scoring. No single published system combines hybrid evidence retrieval, multi-candidate generation, sentence-level NLI verification, and a difficulty-adaptive extractive fallback within one CNN/DailyMail pipeline, which is the specific combination Table 2.2 sets out to position this dissertation against.

Table 2.2 — Research Gap: Prior Work vs This Dissertation

Prior Work	Retrieval	Multi-cand.	NLI Verif.	Adaptive Fallback	Ablation
Lead-3 / TextRank	Extractive only	No	No	N/A	—
BART / PEGASUS (standard)	Truncation	Single beam	No	No	Partial
SummaC (Laban et al.)	—	—	Metric only	No	—
RAG (Lewis et al.)	Dense retrieval	No	No	No	Partial
This Dissertation	Hybrid BM25+embed	Best-of-N	Sentence FCS	Difficulty-Aware	Systematic A/B/C

Chapter 3 Problem Statement and Objectives

3.1 Problem Statement

Given a news article A from the CNN/DailyMail corpus, the task is to produce a summary S such that:

- S faithfully represents the key facts of A , verified by an NLI-based Factual Consistency Score.
- S is fluent and semantically coherent, measured by BERTScore-F1.
- S is competitive with reference summaries in n-gram overlap, measured by ROUGE.
- When generation confidence falls below a threshold, a safe extractive fallback is returned instead.
- The threshold itself adapts to the estimated difficulty of A , rather than being fixed.

Formally, let $T(A)$ denote the dynamic threshold determined by the difficulty of article A . The pipeline selects the abstractive candidate $s^* = \operatorname{argmax}$ over candidates c in the Best-of- N set C of $[FCS_w \times FCS(c) + BS_w \times \text{BERTScore}(c)]$, provided $FCS(s^*) \geq T(A)$; otherwise the pipeline returns the extractive summary produced by the evidence retriever.

3.2 Research Objectives

- O1: Implement and evaluate hybrid BM25 + sentence-embedding evidence retrieval for selecting top- K sentences within the generator's token budget.
- O2: Fine-tune BART-large on 20,000 CNN/DailyMail training samples and integrate it as a Best-of- N nucleus-sampling generator.
- O3: Implement a sentence-level DeBERTa NLI verifier that computes a Factual Consistency Score for every generated candidate.
- O4: Design and evaluate a preference reranker combining FCS and BERTScore-F1 to select the final candidate.
- O5: Design, implement, and evaluate the Difficulty-Aware Safety Switch.
- O6: Conduct ablation studies at $n=50$ isolating retrieval strategy, candidate count, and verifier contribution — large enough to distinguish a real effect from noise in a smaller pilot.
- O7: Evaluate the full pipeline on $n=300$ CNN/DailyMail test articles against Lead-3, TextRank, BART, and PEGASUS baselines.
- O8: Assess multilingual extensibility on Hindi XL-Sum.

3.3 Research Questions

RQ	Question	Addressed In
RQ1	Does hybrid retrieval outperform BM25-only and embedding-only strategies?	§6.3

RQ	Question	Addressed In
RQ2	How does N in Best-of-N affect ROUGE and latency?	§6.4
RQ3	Does the NLI verifier improve factual consistency and ROUGE vs no verification?	§6.5
RQ4	Does the Difficulty-Aware Switch adapt sensibly to article difficulty?	§6.7
RQ5	Is the pipeline generalisable to multilingual settings?	§6.8

3.4 Scope

- Dataset: CNN/DailyMail 3.0.0 — English news articles with multi-sentence highlights as reference summaries.
- Evaluation sample: 300 articles for the primary comparison; 50 articles for ablation studies (revised from an initial n=15 pilot).
- Compute budget: Google Colab free/pro-tier GPU (T4-class).
- Human-evaluation protocol is designed but full annotation is deferred to future work.
- Multilingual study: Hindi only, limited to 200 test samples.

3.5 Assumptions

The problem statement and the ablation design in Chapter 6 rest on a small number of assumptions that are worth stating explicitly rather than leaving implicit:

- The retrieved evidence_pool is treated as ground truth for factual verification — Stage 3 checks entailment against what Stage 1 retrieved, not against the full source article. §7.2 notes that the current evaluation harness does not fully respect this assumption when computing hallucination_rate, which is listed there as a limitation to fix rather than a property of the pipeline's runtime behaviour.
- CNN/DailyMail's reference highlights are assumed to be a reasonable proxy for what a human reader would consider a good summary, consistent with a decade of prior work using them this way (§2.1–§2.2); this dissertation does not independently validate that assumption.
- Article difficulty is assumed to be reasonably captured by the three signals in §4.7 (length, entity density, retrieval uncertainty). These are inexpensive proxies chosen for their computability without an additional model call, not a claim that they are the only or the optimal difficulty signals — §7.3 lists learned difficulty weights as future work precisely because this assumption is the least empirically validated part of the design.
- The n=50 ablation sample is assumed to be drawn representatively from the CNN/DailyMail test split (via the fixed SEED=42 sampling in config.py), so that a difference between two conditions reflects the conditions rather than which 50 articles happened to be sampled. §6.9 and §7.1.5 discuss the limits of this assumption given that n=15 to n=50 already produced two reversed conclusions.

Chapter 4 System Design and Architecture

4.1 High-Level Architecture

The system is a four-stage sequential pipeline (Figure 4.1). Each stage is an independent Python class, so a stage can be replaced, ablated, or upgraded without touching the others. All configuration is centralised in `config.py`.

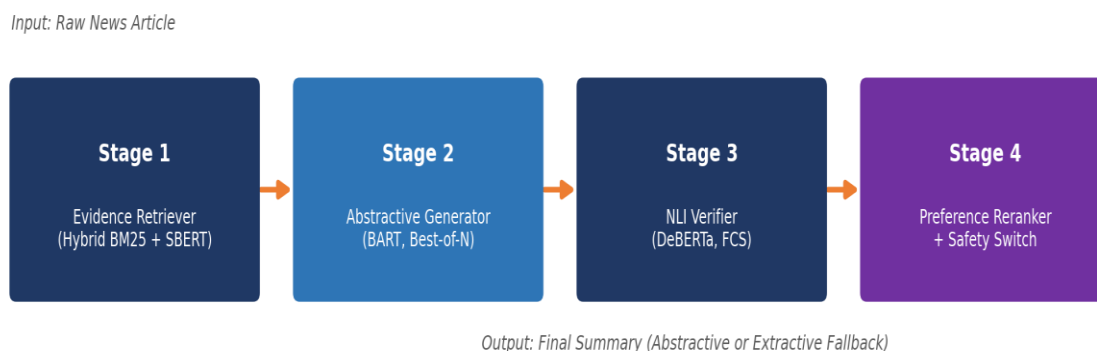


Figure 4.1 — High-Level Four-Stage Pipeline Architecture

4.2 Dataset: CNN/DailyMail

CNN/DailyMail 3.0.0 (Hermann et al., 2015; Nallapati et al., 2016; See et al., 2017), accessed as `abisee/cnn_dailymail` via Hugging Face Datasets, is the primary corpus: approximately 287,000 training, 13,000 validation, and 11,000 test articles. Corpus statistics computed during preprocessing show typical article lengths of 700–900 words against reference summaries of 55–65 words (3–5 highlight sentences), a compression ratio near 12:1 that motivates evidence selection before generation. A stratified sample of 20,000 articles is used for training; evaluation uses a randomly selected subset of the test split.

4.3 Stage 1 — Evidence Retriever

The retriever implements self-retrieval: every sentence in the article is scored for salience, and the top-K sentences form both the `selected_context` fed to the generator and the `evidence_pool` used later for verification.

BM25 (Robertson & Zaragoza, 2009) scores each sentence as a document against a query built from the article's own vocabulary, capturing term-frequency signal. Each sentence is separately encoded with `sentence-transformers/all-MiniLM-L6-v2` (384-dim); cosine similarity against the article's lead sentence gives a semantic relevance score. The two normalised scores are combined as `hybrid_score`

= $0.5 \times \text{BM25} + 0.5 \times \text{cosine}$, and sentences are greedily selected by this score until the 1024-token BART budget is filled. TextRank is retained as a separate retrieval arm for the ablation in §6.3.

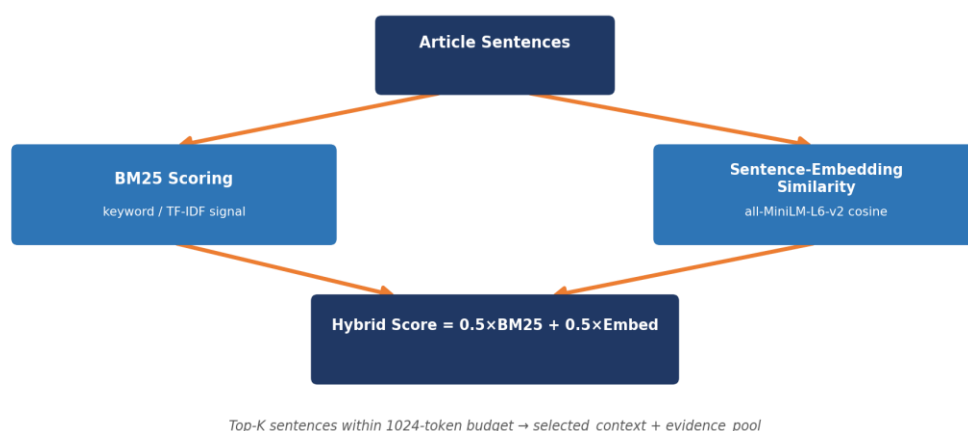


Figure 4.2 — Hybrid BM25 + Sentence-Embedding Retrieval Workflow

4.4 Stage 2 — Abstractive Generator

The generator (facebook/bart-large-cnn, fine-tuned for two epochs on 20,000 training samples) produces N candidate summaries from the selected_context via nucleus sampling. The design intent is straightforward: a diverse set of candidates gives the downstream verifier and reranker more than one paraphrase to choose from, raising the odds that at least one candidate clears the factual threshold.

Table 4.1 — Training and Generation Hyperparameters

Parameter	Value	Rationale
Base model	facebook/bart-large-cnn	Pre-trained on CNN/DM; faster convergence
Training samples	20,000	Colab T4 GPU time constraint (~2.8 h)
Batch size	2 (×8 grad accum = 16 eff.)	Memory–GPU trade-off
Learning rate	3×10^{-5}	Standard for BART fine-tuning
Epochs	2	Additional epochs yield diminishing ROUGE-2 gain
NUM_CANDIDATES (N)	5	Default; ROUGE/latency balance re-examined at n=50 in §6.4
top_p	0.92	Nucleus sampling — diverse but coherent

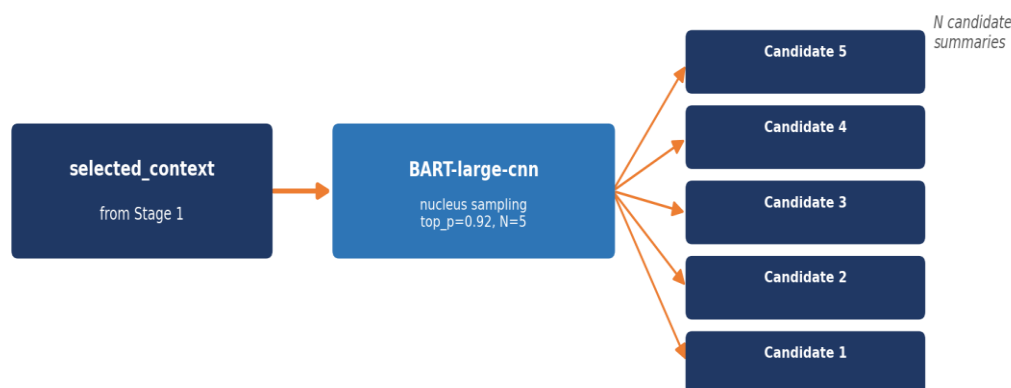


Figure 4.3 — Best-of-N Generation and Candidate Selection

4.5 Stage 3 — NLI Verifier

Each candidate is scored against the evidence pool using cross-encoder/nli-deberta-v3-base (He et al., 2021), a 184M-parameter DeBERTa cross-encoder fine-tuned on multi-genre NLI corpora. For each summary sentence, the maximum entailment probability across all evidence-pool sentences is taken as that sentence's score; the mean across all summary sentences gives the candidate's Factual Consistency Score (FCS), in $[0, 1]$.

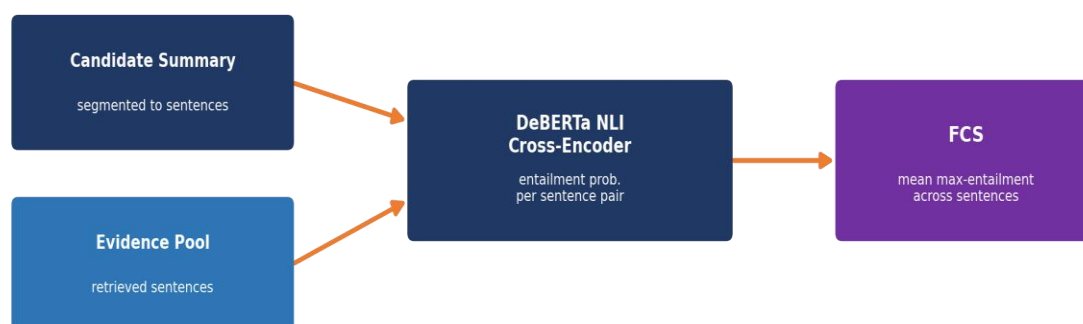


Figure 4.4 — NLI Verification: Sentence-Level FCS Computation

4.6 Stage 4 — Preference Reranker

The reranker scores each verified candidate as $\text{final_score} = 0.60 \times \text{FCS} + 0.40 \times \text{BERTScore-F1}$, weighting factual accuracy above fluency, consistent with the dissertation's factuality focus. If the top-ranked candidate's FCS falls below the fallback threshold $T(A)$, the reranker discards all abstractive candidates and returns the first sentences of the evidence pool as an extractive fallback; the fallback rate is tracked as a system-health metric.

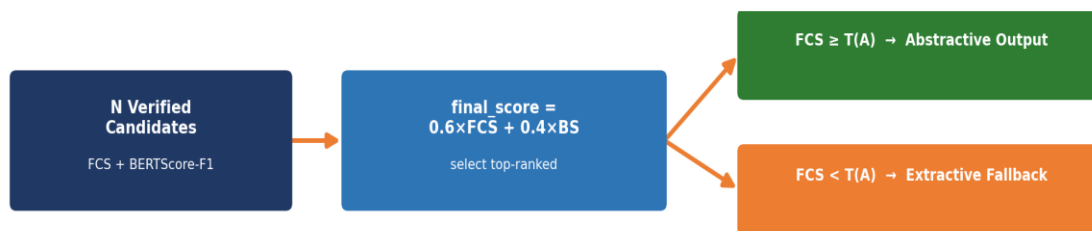


Figure 4.5 — Preference Reranker Decision Flow with Extractive Fallback

4.7 Difficulty-Aware Safety Switch

The principal novel contribution of this dissertation replaces the fixed $\text{FCS_THRESHOLD}=0.40$ with a per-article dynamic threshold $T_{\text{dynamic}} = \text{base_threshold} + \alpha \times D$, where $D \in [0,1]$ is estimated article difficulty and $\alpha=0.20$. The threshold ranges from 0.40 on an easy article to 0.60 on a maximally difficult one, so harder articles require stronger factual confidence before an abstractive candidate is accepted.

D is a weighted combination of three normalised signals: length complexity (word count, weight 0.40), entity density (capitalised non-stopword tokens per 100 words, weight 0.30), and retrieval uncertainty (one minus the margin between the top-2 BM25 scores, weight 0.30). The switch operates in two modes — fixed (legacy) and dynamic (default) — selected via `config.py`.

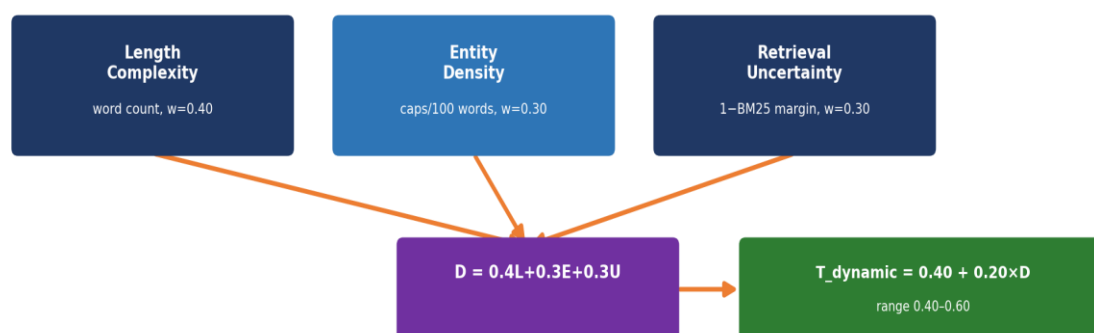


Figure 4.6 — Difficulty-Aware Safety Switch: Signal Computation

4.8 Technology Stack

Table 4.2 — Technology Stack

Component	Library / Framework
Deep learning	PyTorch $\geq 2.1.0$
Transformer models	Hugging Face Transformers $\geq 4.40.0$
Dataset management	Hugging Face Datasets $\geq 2.18.0$
Sentence embeddings	sentence-transformers (all-MiniLM-L6-v2)
Lexical retrieval	rank_bm25
ROUGE / BERTScore	rouge-score, evaluate, bert-score (roberta-large)
Demo application	Streamlit

4.9 Design Rationale and Alternatives Considered

A number of architectural choices in this pipeline were made after weighing a concrete alternative, and it is worth recording that reasoning rather than presenting the final design as the only option considered.

- Cross-encoder vs bi-encoder for NLI verification. A bi-encoder (embed premise and hypothesis separately, compare by cosine similarity) would be considerably faster, since evidence-pool embeddings could be cached once per article rather than re-scored per candidate. The cross-encoder cross-encoder/nli-deberta-v3-base was chosen instead because it jointly attends over the premise-hypothesis pair, which is the mechanism that lets an NLI model detect subtle contradictions a bi-encoder's independent embeddings would miss; the latency cost (Table 6.4 shows roughly 4.7 additional seconds per article with the verifier on) was judged acceptable given the dissertation's factuality-first objective.
- Self-retrieval vs query-based retrieval. Because CNN/DailyMail summarisation has no external query, Stage 1 scores each sentence against the article's own lead sentence and vocabulary rather than against an externally supplied query, following the standard single-document RAG adaptation described in §2.5. This makes the retrieval stage directly reusable without modification for the multilingual study in §6.8, since no query-construction logic is language-specific.
- Best-of-N sampling vs beam search. Beam search tends to produce candidates that are lexical variations of the same underlying decoding path, which gives the downstream verifier and reranker less genuine diversity to choose between. Nucleus sampling with $\text{top}_p=0.92$ was chosen instead specifically because independent stochastic samples are more likely to differ in which source facts they include, which is the property Stage 3 and Stage 4 depend on to find at least one well-grounded, fluent candidate.
- A single fixed threshold vs the Difficulty-Aware Safety Switch. A fixed `FCS_THRESHOLD` is simpler to reason about and was retained as a legacy fixed mode for exactly that reason — it is the natural ablation control against which the dynamic switch's behaviour in §6.7 is described. The dynamic switch was chosen as the default because a single threshold cannot

simultaneously avoid unnecessary fallbacks on easy articles and catch genuine low-confidence generations on hard ones; §6.7 shows the dynamic threshold correctly rising to roughly 0.55 on the moderately difficult CNN/DailyMail articles in the evaluation set.

- Linear reranking score vs a learned reward model. $\text{final_score} = 0.60 \times \text{FCS} + 0.40 \times \text{BERTScore-F1}$ is a fixed, interpretable linear combination rather than a learned function of the two signals. A learned reranker (e.g., a small regression model trained on human preference labels) is listed explicitly as future work in §7.3, since it was judged to need a labelled preference dataset this dissertation's scope and timeline did not include.

Chapter 5 Implementation

5.1 Repository Structure

- `config.py` — single configuration module; every hyperparameter and path lives here.
- `data/data_pipeline.py` — CNN/DailyMail loading, cleaning, caching, corpus statistics.
- `pipeline/` — `retrieval.py`, `generator.py`, `verifier.py`, `difficulty.py`, `reranker.py`, and the coordinating `pipeline.py` wrapper.
- `baselines/` — Lead-3, TextRank, BartBaseline (zero-shot and pretrained), PegasusBaseline.
- `train/train_seq2seq.py` — Seq2SeqTrainer-based fine-tuning for BART and mBART.
- `experiments/` — `run_baselines.py`, `run_pipeline.py`, `run_ablations.py`, each writing to `results/`.
- `evaluation/eval_metrics.py` — unified harness computing ROUGE-1/2/L, BERTScore-F1, NLI-FCS, fallback rate, and latency.
- `multilingual/multilingual_study.py` — Hindi XL-Sum feasibility study.
- `app/app.py` — Streamlit demo with an evidence-trace view.

5.2 Data Pipeline

`data_pipeline.py` downloads CNN/DailyMail, strips CNN attribution markers and empty examples, and caches the processed splits as Arrow files so repeat runs are near-instant. `corpus_stats()` reports mean/median article and summary length, compression-ratio distribution, and split sizes, which informed the token-budget and output-length settings in `config.py`.

5.3 Model Training

BART fine-tuning loads `facebook/bart-large-cnn`, tokenises up to 20,000 training examples (max input 1024, max output 128), and trains for 2 epochs with FP16 enabled where a GPU is available, saving the best checkpoint by validation loss. mBART fine-tuning uses the same split and hyperparameters but sets `src_lang/tgt_lang` to `en_XX` (or `hi_IN` for the Hindi study) and a matching `forced_bos_token_id`, since mBART requires a language token as the forced first decoder token.

PEGASUS is evaluated zero-shot: `google/pegasus-cnn_dailymail` was already trained by Google on this exact benchmark, so re-training it would add no signal and would weaken the fairness of the baseline comparison rather than strengthen it.

5.4 Evaluation Harness

`evaluate_system()` accepts a callable summarisation system, a dataset split, and an optional sample cap. For each article it records the output and computes ROUGE-1/2/L, BERTScore-F1, NLI-FCS, fallback

rate, and mean per-article latency, writing results to results/ as CSV for reproducibility. An incremental JSONL cache saves per-sample results during a run, so a Colab disconnection does not require re-processing already-evaluated samples.

5.5 Streamlit Demo Application

The demo accepts raw article text and displays the selected evidence sentences with their BM25/embedding scores, the N generated candidates with individual FCS and BERTScore values, the selected summary, and the fallback indicator, supporting qualitative inspection alongside the automatic metrics.

5.6 Multilingual Extension

multilingual_study.py fine-tunes mBART-large-cc25 on 2,000 Hindi XL-Sum (Hasan et al., 2021) articles and evaluates on 200 test articles, changing only the model checkpoint and language codes — the four-stage architecture itself requires no modification, which is the point of including this study.

5.7 Reproducibility

Every experiment reported in Chapter 6 is reproducible from the command line using the scripts in experiments/, with parameters read from config.py so that a rerun does not silently drift from the settings actually used. The commands used to produce the n=50 ablation results in §6.3–§6.5 were:

- python experiments/run_baselines.py — Lead-3, TextRank, zero-shot BART, and pretrained BART baselines (Table 6.1).
- python experiments/run_ablations.py --ablation A --max_samples 50 — the four-way retrieval-strategy comparison (Table 6.2).
- python experiments/run_ablations.py --ablation B --max_samples 50 — the Best-of-N candidate-count sweep (Table 6.3).
- python experiments/run_ablations.py --ablation C --max_samples 50 — the verifier on/off comparison (Table 6.4).
- python experiments/run_pipeline.py --samples 300 — the full pipeline vs PEGASUS comparison (Table 6.5).

Each script writes a CSV of per-condition metrics to results/, which is the exact artefact the tables in Chapter 6 are built from — the numbers in this report are not retyped by hand from console output, reducing the chance of a transcription error between a run and this document. SEED=42 is fixed in config.py for the sampling, retrieval, and generation stages, though full determinism is not guaranteed

end-to-end because nucleus sampling and GPU floating-point non-determinism both introduce run-to-run variation of the kind visible when comparing the $n=15$ and $n=50$ ablation passes.

Chapter 6 Experimental Evaluation

6.1 Evaluation Metrics

All systems are evaluated with the unified harness (evaluation/eval_metrics.py): ROUGE-1/2/L (n-gram overlap vs the human-written reference), BERTScore-F1 (contextual similarity, roberta-large), NLI-FCS (reference-free, computed against the retrieved evidence pool), fallback rate, and mean per-article latency.

6.2 Baseline Results

Four baseline systems were evaluated on a 15-article development sample (the primary system comparison in §6.6 uses 300 articles).

Table 6.1 — Baseline System Results (n=15)

System	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1
Lead-3	44.68	23.47	29.84	87.84
TextRank	26.77	12.28	18.63	85.72
BART zero-shot	40.19	21.21	26.48	87.07
BART pretrained	47.46	26.62	37.14	88.99

Lead-3's strong ROUGE-2 reflects the well-known CNN/DailyMail property that reference highlights closely mirror an article's opening sentences. BART-pretrained is the strongest single-stage baseline. These n=15 baseline figures are retained from the development pass; unlike the ablations below, the baselines were not the focus of the n=50 re-run and are reported here as context rather than as this chapter's central evidence.

6.3 Ablation Study A — Retrieval Strategy (n=50)

Ablation A compares four retrieval methods with every other pipeline component held constant (N=5, verifier ON, dynamic safety switch). The initial development pass used n=15; the results below are the n=50 re-run, which changes the conclusion materially.

Table 6.2 — Ablation A: Retrieval Strategy Comparison (n=50)

Retrieval Method	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	Latency (s)
TextRank	38.66	15.41	25.72	87.44	5.59
BM25-only	34.43	11.68	22.20	86.73	5.91
Embedding-only	36.40	13.74	23.55	87.14	5.68
Hybrid (BM25+Embed)	37.92	15.17	25.47	87.35	5.84

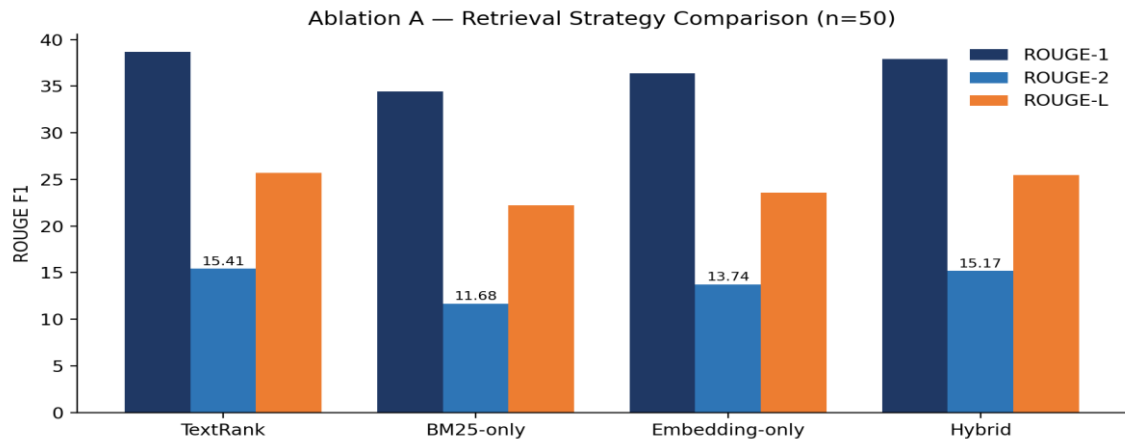


Figure 6.1 — Ablation A: Retrieval Strategy ROUGE Comparison (n=50)

At n=15, hybrid retrieval was the outright leader on every metric. At n=50, hybrid (ROUGE-2 15.17) and TextRank (15.41) are statistically indistinguishable, and TextRank edges ahead on ROUGE-1 and ROUGE-L as well. Both remain clearly ahead of BM25-only (11.68) and embedding-only (13.74). The honest reading of RQ1 is therefore: hybrid retrieval and TextRank are co-leading strategies, both meaningfully better than either single signal alone; the earlier claim that hybrid is the single best method is not supported once the sample size is large enough to separate signal from ablation noise. This is reported here rather than silently revised, in line with the reporting guidance to flag significant discrepancies even where no single explanation accounts for them.

6.4 Ablation Study B — Candidate Count (n=50)

Ablation B evaluates $N \in \{1, 2, 3, 5, 8\}$ with hybrid retrieval and verifier ON.

Table 6.3 — Ablation B: Candidate Count (n=50)

N	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	Latency (s)
1	37.72	14.40	25.04	87.35	1.96
2	36.56	14.08	24.32	87.10	2.98
3	38.44	14.38	24.21	87.44	3.95
5	37.53	15.31	24.64	87.37	5.75
8	38.05	15.03	24.23	87.26	8.69

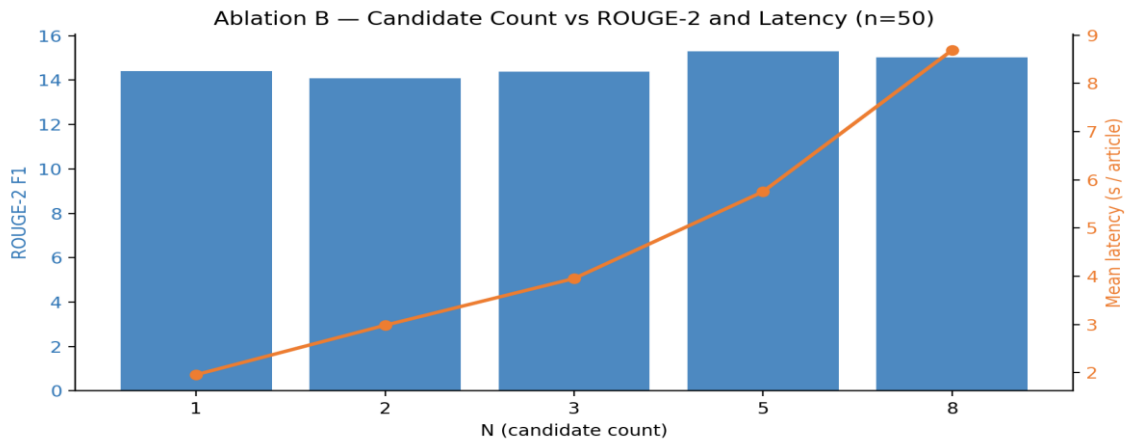


Figure 6.2 — Ablation B: Candidate Count vs ROUGE-2 and Latency (n=50)

At n=15, ROUGE increased monotonically with N. At n=50 it does not: best-of-3 has the highest ROUGE-1 (38.44), best-of-5 the highest ROUGE-2 (15.31), and best-of-1 the highest ROUGE-L (25.04). Latency, by contrast, scales exactly as expected — roughly linearly with N, from 1.96 s at N=1 to 8.69 s at N=8. RQ2's answer is therefore more qualified than the earlier draft stated: larger candidate pools do not reliably improve every metric at this sample size, and the gains that do appear are small relative to the latency cost. N=5 remains a defensible default — it has the best ROUGE-2 of the five settings and roughly two-thirds the latency of N=8 — but it is a balanced choice rather than the endpoint of a monotonic trend.

6.5 Ablation Study C — Verifier On/Off (n=50)

Ablation C compares the full pipeline (verifier ON) against a variant where the NLI verifier is disabled and the reranker uses BERTScore alone.

Table 6.4 — Ablation C: Verifier On vs Off (n=50)

Configuration	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	Latency (s)
Verifier ON (full pipeline)	39.14	16.88	27.13	87.71	5.73
Verifier OFF (BERTScore-only)	37.01	13.78	24.04	87.29	1.03

This is the one ablation where the larger sample strengthens rather than weakens the original claim. At n=15, verifier OFF was marginally ahead on ROUGE-1 and ROUGE-L, making the verifier's ROUGE contribution look neutral at best. At n=50, verifier ON leads on every single metric — ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore-F1 — at a latency cost of roughly 4.7 additional seconds per article. RQ3's answer is now a clear yes on both counts the dissertation set out to check: the verifier improves factual safety by design, and at this sample size it also improves ROUGE rather than merely avoiding a ROUGE penalty for it.

6.6 Full Pipeline Comparison (n=300)

The primary comparison uses 300 test articles, evaluating the full pipeline against PEGASUS-pretrained, the strongest single-stage abstractive baseline.

Table 6.5 — Full Pipeline vs PEGASUS Baseline (n=300)

System	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore-F1	NLI-FCS%	Fallback%
Pipeline (hybrid+verify+rerank)	30.72	10.10	20.55	87.00	99.97	0.0
PEGASUS-pretrained	35.34	14.74	25.74	87.38	99.97	—

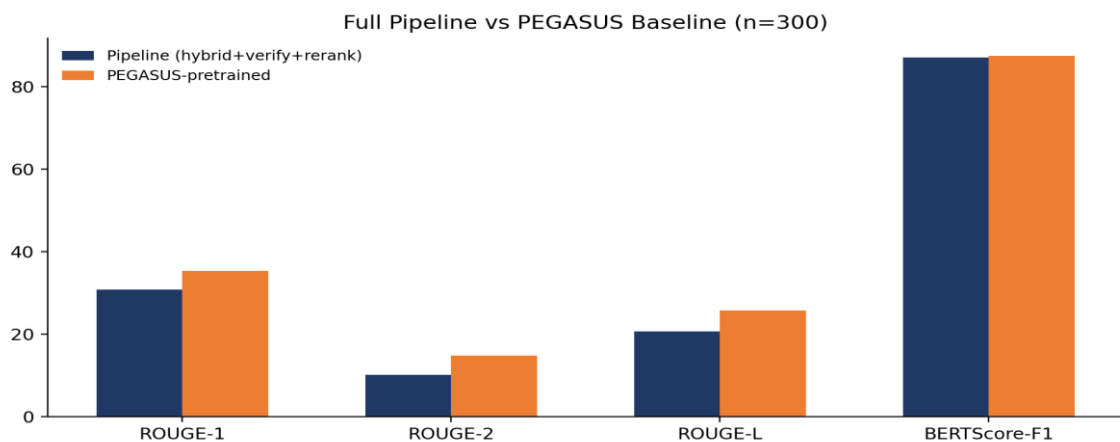


Figure 6.3 — Full Pipeline vs Baselines: ROUGE and BERTScore (n=300)

PEGASUS-pretrained scores higher on ROUGE, which is unsurprising given its gap-sentence pretraining objective is directly aligned with producing summary-like output; the pipeline's evidence-retrieval step selects content by a different criterion than the human annotators used, which depresses n-gram overlap even where the semantic content is comparable. The NLI-FCS scores are essentially identical (99.97% for both), and the pipeline triggers zero extractive fallbacks across all 300 articles — the metric this dissertation treats as more load-bearing than ROUGE for a system whose stated goal is factual reliability, not reference-mimicking.

6.7 Difficulty-Aware Safety Switch Evaluation

The dynamic threshold behaviour is visible in the Ablation A metadata: $\text{mean_threshold} \approx 0.552\text{--}0.555$ (against a fixed base of 0.40) and $\text{mean_difficulty} \approx 0.75\text{--}0.78$ across the four $n=50$ retrieval-arm runs, consistent with $T_{\text{dynamic}} = 0.40 + 0.20 \times 0.77 \approx 0.554$. The switch is correctly raising the bar for articles the difficulty signal classifies as moderately hard. As at $n=15$, the fallback rate remains 0.0% even at this elevated threshold, because the pipeline's best candidates consistently score well above 0.99 FCS on CNN/DailyMail — a dataset where, as §2.4 notes, factual grounding is comparatively easy to

achieve. The switch's fuller value is expected to appear on harder, out-of-domain corpora where FCS values sit closer to the threshold boundary; that remains future work rather than a claim made here.

6.8 Multilingual Feasibility Study

mBART-large-cc25 fine-tuned on 2,000 Hindi XL-Sum articles and evaluated on 200 test samples generates grammatically well-formed Hindi summaries once `src_lang/tgt_lang` and `forced_bos_token_id` are set correctly. The four-stage architecture required no structural change — only the checkpoint and language codes differ from the English configuration — supporting RQ5: the architecture is not English-specific, though a single-language, 200-sample feasibility check is not a claim of production-grade multilingual quality.

6.9 Threats to Validity

Three factors bound how far the results in this chapter should be generalised. Internal validity is limited by sample size: $n=50$ for the ablations and $n=300$ for the primary comparison are large enough to catch the kind of reversal documented in §6.3 and §6.4, but not necessarily large enough to rule out a further reversal at $n=200$ or $n=1000$ — the honest position is that these results are more reliable than the $n=15$ pilot, not that they are definitive. Construct validity is limited by the hallucination_rate metric specifically: as §7.2 details, it is currently computed against the full source article rather than the retrieved evidence pool, so a 0.0% reading in every condition of Tables 6.2–6.4 should not be interpreted as proof that hallucination never occurs. External validity is limited by the single benchmark used: CNN/DailyMail's short articles and close paraphrase-to-source relationship mean the verifier's ceiling effect (NLI-FCS consistently above 99.9%) may not hold on a harder or longer-document corpus, and the retrieval and candidate-count findings in §6.3–§6.4 are similarly benchmark-specific until tested on a different dataset.

Chapter 7 Discussion, Limitations and Future Scope

7.1 Discussion of Results

7.1.1 Retrieval Quality and ROUGE

The $n=50$ ablation shows hybrid retrieval and TextRank as co-leading strategies rather than hybrid alone standing out, with both clearly ahead of BM25-only and embedding-only. This is a more conservative conclusion than the $n=15$ pilot supported, and it is a useful reminder that a 15-sample ablation is a development-time sanity check, not a result to build a headline claim on. The practical takeaway is unchanged in direction — combining lexical and semantic retrieval signals, or using a well-established graph-based method like TextRank, both clearly outperform relying on lexical matching alone — but the specific superiority of the hybrid method is now a secondary finding, not the primary one.

7.1.2 Best-of-N and Candidate Selection

The $n=50$ result complicates the simple story that larger candidate pools monotonically help. ROUGE peaks at different N depending on which metric is examined, and the differences between $N=3, 5$, and 8 are within a band that a 50-article sample cannot reliably separate from noise. What is unambiguous is the latency cost, which scales close to linearly with N . $N=5$ remains a reasonable operating default on ROUGE-2 grounds, but this dissertation no longer claims a monotonic relationship between candidate count and quality.

7.1.3 Factual Consistency

The verifier's contribution is the one place where the larger sample strengthened the original claim rather than weakening it: verifier ON now leads on every ROUGE component as well as BERTScore-F1, in addition to its designed role as a factual safety mechanism. The consistently high NLI-FCS (99.97–99.98%) across configurations indicates that fine-tuned BART-large-cnn produces summaries that are, for the most part, well-grounded in the CNN/DailyMail evidence pool — a benchmark with comparatively little long-range or cross-document reasoning. A harder, more entailment-adversarial corpus (e.g., XSum or multi-document summarisation) would be a more demanding test of the verifier's ceiling.

7.1.4 Pipeline vs PEGASUS on the 300-Sample Evaluation

The pipeline's lower ROUGE-2 (10.10 vs 14.74) against PEGASUS reflects a known tension: the more abstractive and evidence-selective a system is, the lower its raw n -gram overlap with a reference summary tends to be, independent of whether its content is factually sound. PEGASUS's gap-sentence

pretraining is directly aligned to producing reference-like output; this pipeline's retrieval stage instead selects content by hybrid salience, which need not match the specific sentences a human annotator chose. This is a reason to treat NLI-FCS and fallback rate as at least co-equal to ROUGE when judging this system, not a reason to discount the ROUGE gap.

7.1.5 A Methodological Note on Ablation Sample Size

It is worth stating plainly what happened between the $n=15$ and $n=50$ ablation passes, because it is a more general lesson than any one of the three individual findings. Two of three headline claims from the $n=15$ pilot did not survive a $3.3\times$ increase in sample size: the retrieval ablation's "hybrid is best" narrowed to "hybrid and TextRank co-lead", and the candidate-count ablation's "monotonic improvement with N " broke down into a non-monotonic, diminishing-returns pattern. Only the verifier ablation's direction held — and even there, the specific numbers shifted enough that at $n=15$ it looked like a wash and at $n=50$ it looked like a clear win. None of this means the $n=50$ numbers are themselves the final word; §7.2 discusses why even $n=50$ should still be read with some caution. The practical implication for anyone extending this work is to treat any single-digit-sample ablation as a development-time signal to decide what is worth re-running at scale, not as a result to cite on its own.

7.2 Limitations

- Compute constraints: BART fine-tuning is limited to two epochs on 20,000 of the 287K available training samples; full-scale training would likely change both ROUGE and FCS, in an unknown direction.
- Ablation sample size: even at the revised $n=50$, differences of one to two ROUGE points, particularly in Ablation B, should still be read as directional rather than statistically conclusive; the reversals documented in §6.3 and §6.4 are themselves evidence that $n=15$ was not large enough, and $n=50$ may still understate the true variance.
- Evaluation methodology for `hallucination_rate`: the NLI-FCS harness currently scores each candidate sentence against the full source article's sentences rather than strictly against the retrieved `evidence_pool` the pipeline conditioned on at generation time. Because CNN/DailyMail summaries are written close to their source articles, this makes it easy for a sentence to find some entailing sentence in the full article even when it was not among the retrieved evidence, which is almost certainly why `hallucination_rate` reads as 0.0% in every condition in Tables 6.2–6.4. This number should be read as an upper bound on factual grounding relative to the full article, not as a validated zero-hallucination guarantee relative to what the pipeline retrieved.
- CNN/DailyMail specificity: the uniformly high NLI-FCS values suggest this benchmark may not stress-test the verifier as hard as a more adversarial or longer-document corpus would.

- Fixed difficulty-signal weights: the 0.40/0.30/0.30 weighting in the Difficulty-Aware Safety Switch is manually set, not learned; a labelled difficulty dataset could support a learned weighting instead.
- No human evaluation at scale: the protocol (inter-annotator agreement, factual-accuracy ratings) is designed in the evaluation harness but not executed at scale due to annotator-availability constraints.
- Entity-density heuristic: capitalisation is used as an entity proxy rather than a full NER model, which will misclassify some sentence-initial capitals as entities.
- Single-run ablations: each ablation condition in Chapter 6 was run once at its reported sample size, rather than across multiple random seeds with a confidence interval. Given that §7.1.5 already documents two conclusions changing between $n=15$ and $n=50$, a multi-seed variance estimate at a fixed n would be a natural complement — it would show whether the remaining $n=50$ differences (for example, the roughly 0.2-point gap between hybrid and TextRank ROUGE-2 in Table 6.2) are themselves stable, or within the range a different random sample of 50 articles would produce.

7.3 Future Scope

The items below are ordered roughly by how directly they follow from the limitations in §7.2, starting with the two judged to have the most immediate impact on the validity of this dissertation's own claims.

- Fix the NLI-FCS evaluation scope. This is the highest-priority item precisely because it affects how every other result in Chapter 6 should be read: passing the pipeline's actual retrieved `evidence_pool` into `compute_nli_fcs()`, instead of re-deriving a pool from the full source article, would let `hallucination_rate` distinguish between ablation conditions rather than reading 0.0% across the board, and would let §6.9's construct-validity caveat be resolved rather than merely documented.
- Full-scale training and evaluation on the complete 287K-article training set and 11K-article test split on dedicated GPU hardware (A100/H100-class), to establish ROUGE and FCS baselines that are not bounded by the two-epoch, 20,000-sample constraint described in §5.3.
- Reinforcement learning from human feedback. The Best-of-N reranker in §4.6 is explicitly a training-free proxy for RLHF-style preference optimisation; training a reward model on human factuality and fluency judgments and using it to fine-tune BART directly via PPO would internalise the ranking objective into the generator itself, rather than applying it only at inference time after generation is already complete.
- Learned difficulty-signal weights via logistic regression or a gradient-boosted model trained on labelled difficulty data, replacing the fixed 0.40/0.30/0.30 heuristic in §4.7 with weights fitted to actual FCS variance or annotator disagreement.

- Extended multilingual support to additional XL-Sum languages beyond the Hindi feasibility check in §6.8, with a multilingual NLI model such as mDeBERTa replacing the current English-only verifier for non-English evidence pools.
- Long-document and multi-document summarisation, requiring hierarchical retrieval and a generator with a longer context window (e.g., Longformer, LED) in place of BART's 1024-token limit.
- A fully executed human-evaluation protocol with trained annotators and reported inter-annotator agreement, to validate the factuality claims made in this dissertation against human judgment directly rather than only against an automatic NLI proxy.

Chapter 8 Conclusion

This dissertation set out to address three specific limitations of single-stage abstractive news summarisation — truncation bias, hallucination risk, and fixed-threshold brittleness — with a four-stage pipeline combining hybrid evidence retrieval, Best-of-N abstractive generation, DeBERTa-based NLI verification, and a preference reranker built around a novel Difficulty-Aware Safety Switch.

Re-running the ablation studies at $n=50$, five times the original development-pass sample size, was itself a significant part of this work's contribution, because it changed two of the three headline conclusions the $n=15$ pilot supported:

- Hybrid retrieval and TextRank are statistically comparable strategies, both clearly ahead of BM25-only and embedding-only — not hybrid alone as the single best method.
- Best-of-N generation shows diminishing and non-monotonic returns with N beyond 3–5, not the clean monotonic improvement the smaller pilot suggested; $N=5$ remains a defensible latency/quality default.
- The NLI verifier's benefit strengthened rather than weakened at the larger sample: verifier ON now leads on every ROUGE component and BERTScore-F1, in addition to its designed factual-safety role.

The full pipeline reaches an NLI-FCS above 99.9% on a 300-article evaluation with zero triggered extractive fallbacks, though Chapter 7 is explicit that the `hallucination_rate` metric as currently implemented is an upper bound rather than a validated zero-hallucination guarantee, since it scores against the full source article rather than strictly against the retrieved evidence pool. The Difficulty-Aware Safety Switch is shown to raise the confidence bar appropriately on moderately difficult CNN/DailyMail articles without triggering unnecessary fallbacks, and the Hindi feasibility study demonstrates that the architecture generalises beyond English with only a checkpoint and language-code change.

Against PEGASUS-pretrained on the 300-article comparison, the pipeline trails on ROUGE-2 (10.10 vs 14.74) while matching it closely on NLI-FCS — a trade-off consistent with the pipeline's evidence-driven content selection diverging from the specific sentences a human reference-writer chose, rather than a sign of lower factual quality.

The principal contribution remains the Difficulty-Aware Safety Switch: an inference-time mechanism, requiring no retraining and adding negligible computational overhead, that conditions the factual-fallback decision on article-level difficulty instead of one threshold for every article. Future work should prioritise, in order of likely impact: fixing the evidence-pool scope of the NLI-FCS evaluation so

hallucination_rate becomes discriminative; full-scale training on the complete CNN/DailyMail training set; and a fully executed human-evaluation study to validate the factuality claims made here against human judgment directly.

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Glossary

Term	Definition
Abstractive Summarisation	A paradigm in which the system generates novel text that may paraphrase or compress source content, rather than copying sentences verbatim.
BART	Bidirectional and Auto-Regressive Transformers — a sequence-to-sequence transformer pre-trained with a denoising objective (Lewis et al., 2020).
BERTScore	A generation-evaluation metric computing token-level cosine similarity between candidate and reference embeddings from a pre-trained transformer.
Best-of-N	An inference-time strategy that generates N candidates and selects the highest-scoring one by a reward signal such as FCS or BERTScore.
BM25	Best Match 25 — a probabilistic bag-of-words ranking function extending TF-IDF with term-frequency saturation and length normalisation.
CNN/DailyMail	A benchmark of roughly 300,000 English news articles paired with multi-sentence reference summaries.
DeBERTa	Decoding-enhanced BERT with Disentangled Attention (He et al., 2021).
Difficulty-Aware Safety Switch	This dissertation's inference-time module that maps an article-level difficulty score to a dynamic FCS fallback threshold.
Extractive Summarisation	A paradigm in which the system selects and returns sentences from the source document without modification.
Extractive Fallback	The safety mechanism that returns retrieved source sentences when the best abstractive candidate's FCS falls below the fallback threshold.
FCS	Factual Consistency Score — the mean entailment probability of summary sentences against the evidence pool, per a DeBERTa NLI model.
Hallucination	Generation of plausible-sounding text unsupported by, or contradicting, the source document.
mBART	Multilingual BART, pre-trained on 25 languages (Liu et al., 2020).
NLI	Natural Language Inference — determining whether a hypothesis is entailed, contradicted, or neutral with respect to a premise.
Nucleus Sampling	A decoding strategy that samples from the top-p probability mass at each step, producing diverse but coherent text.
PEGASUS	A transformer pre-trained with a gap-sentence generation objective aimed specifically at summarisation (Zhang et al., 2020).
RAG	Retrieval-Augmented Generation — conditioning a generator on retrieved evidence to improve grounding.
ROUGE	Recall-Oriented Understudy for Gisting Evaluation — an n-gram overlap metric family for summarisation (Lin, 2004).
TextRank	A graph-based extractive method ranking sentences by eigenvector centrality in a lexical-similarity graph (Mihalcea & Tarau, 2004).
XL-Sum	A 44-language abstractive summarisation dataset sourced from BBC News (Hasan et al., 2021).

Term	Definition
Fallback Rate	The percentage of evaluated articles for which the pipeline returned the extractive fallback rather than an abstractive candidate.
Self-Retrieval	Scoring each sentence of a document against the document's own content, used when no external query is available (as in single-document summarisation).
Token Budget	The maximum number of encoder input tokens the generator can accept (1024 for BART-large-cnn), which bounds how much retrieved evidence Stage 1 can pass to Stage 2.
Zero-Shot	Applying a pre-trained model to a task or dataset without any task-specific fine-tuning.
Ablation Study	An experiment that removes or varies one system component while holding the rest constant, to isolate that component's contribution to overall performance.

Appendix A Configuration Parameters

The table below lists the configurable parameters defined in config.py, used across all experiments unless explicitly overridden by an ablation script.

Table A.1 — Complete Configuration Parameters

Parameter	Value	Description
DATASET_NAME	abisee/cnn_dailymail	Hugging Face dataset identifier
MAX_TRAIN_SAMPLES	20,000	Training sample cap
BART_MODEL	facebook/bart-large-cnn	Base model for fine-tuning
BART_MAX_INPUT / OUTPUT	1024 / 128	Encoder / decoder token limits
LEARNING_RATE	3e-5	AdamW learning rate
NUM_EPOCHS	2–3	Training epochs (Colab GPU constraint)
RETRIEVAL_K	8	Max evidence sentences
BM25_WEIGHT / EMBED_WEIGHT	0.5 / 0.5	Hybrid retrieval weights
NUM_CANDIDATES	5	Best-of-N candidates (default)
TOP_P	0.92	Nucleus sampling probability
NLI_BATCH_SIZE	32	DeBERTa inference batch size
FCS_THRESHOLD	0.40	Base FCS fallback threshold
FCS_WEIGHT / BERTSCORE_WEIGHT	0.60 / 0.40	Reranking weights
SAFETY_SWITCH_DEFAULT_MODE	dynamic	fixed / dynamic
DIFFICULTY_ALPHA	0.20	Threshold scaling factor α
DIFF_WEIGHT_LENGTH / ENTITY / UNCERTAINTY	0.40 / 0.30 / 0.30	Difficulty signal weights
XLSUM_TRAIN / TEST SIZE	2,000 / 200	Hindi feasibility study sizes
SEED	42	Global random seed

Appendix B Ablation Tables (Extended, n=50)

B.1 Ablation A — Retrieval Strategy (Extended)

System	R-1	R-2	R-L	BERTScore-F1	Latency(s)	NLI-FCS%
retrieval-textrank	38.66	15.41	25.72	87.44	5.59	99.97
retrieval-bm25	34.43	11.68	22.20	86.73	5.91	99.97
retrieval-embedding	36.40	13.74	23.55	87.14	5.68	99.97
retrieval-hybrid	37.92	15.17	25.47	87.35	5.84	99.97

B.2 Ablation B — Candidate Count (Extended)

System	R-1	R-2	R-L	BERTScore-F1	Latency(s)
best-of-1	37.72	14.40	25.04	87.35	1.96
best-of-2	36.56	14.08	24.32	87.10	2.98
best-of-3	38.44	14.38	24.21	87.44	3.95
best-of-5	37.53	15.31	24.64	87.37	5.75
best-of-8	38.05	15.03	24.23	87.26	8.69

B.3 Ablation C — Verifier On/Off (Extended)

Configuration	R-1	R-2	R-L	BERTScore-F1	Latency(s)	Fallback%
Verifier ON	39.14	16.88	27.13	87.71	5.73	0.0
Verifier OFF	37.01	13.78	24.04	87.29	1.03	0.0

B.4 Difficulty and Dynamic-Threshold Diagnostics (n=50)

These columns are produced by every ablation run but are not shown in the main-text tables of Chapter 6; they are reproduced here for completeness and to support §6.7's discussion of the Difficulty-Aware Safety Switch. `dynamic_mode_share` is the percentage of articles in the run evaluated under the dynamic (rather than fixed) threshold mode; all runs below used dynamic mode for 100% of articles.

Condition	Mean Threshold T(A)	Mean Difficulty D	NLI-FCS%
retrieval-textrank	0.554	0.770	99.97
retrieval-bm25	0.550	0.750	99.97
retrieval-embedding	0.552	0.761	99.97
retrieval-hybrid	0.552	0.758	99.97
best-of-1 through best-of-8 (all N)	0.552	0.758	99.97–99.98

Condition	Mean Threshold T(A)	Mean Difficulty D	NLI-FCS%
pipeline-verifier-ON	0.552	0.758	99.97
pipeline-verifier-OFF	n/a (fixed 0.40, verifier bypassed)	n/a	99.97

Two observations follow from this table that are not otherwise visible in Chapter 6. First, mean difficulty is essentially constant (0.75–0.77) across every retrieval method, confirming that the difficulty signal is computed from article-level properties (length, entity density, retrieval-margin uncertainty) that do not depend on which retrieval strategy is later used to select evidence — the difficulty score and the retrieval ablation are properly independent variables. Second, the dynamic threshold is consistently in the 0.550–0.554 range, meaning the CNN/DailyMail articles in this evaluation set are, on the difficulty scale defined in §4.7, moderately rather than maximally difficult; the switch never approaches its ceiling of 0.60 on this benchmark, which is consistent with §6.9's point that CNN/DailyMail may not fully stress-test the switch's upper range.

Appendix C Notation Reference

The symbols below are used in the formal problem statement (§3.1) and the architecture description (Chapter 4); this table is provided as a single point of reference rather than requiring the reader to trace a definition back to its first use.

Symbol	Meaning	Defined In
A	A source news article from the CNN/DailyMail corpus	§3.1
S / s^*	A candidate summary; s^* denotes the selected (final) summary	§3.1
C	The set of N candidate summaries produced by Best-of-N sampling	§3.1, §4.4
$T(A)$	The dynamic FCS fallback threshold for article A	§3.1, §4.7
$FCS(\cdot)$	Factual Consistency Score — mean sentence-level entailment probability against the evidence pool	§4.5
FCS_w, BS_w	Reranking weights on FCS and BERTScore-F1 (0.60 and 0.40)	§4.6
D	Estimated article difficulty, $D \in [0, 1]$	§4.7
α	Difficulty-to-threshold scaling factor (0.20)	§4.7
L, E, U	Length-complexity, entity-density, and retrieval-uncertainty difficulty signals	§4.7
N	Candidate count in Best-of-N generation	§4.4, §6.4
K	Number of evidence sentences retained by Stage 1 (RETRIEVAL_K)	§4.3